

Probability Homework 4

B13902022 LAI, YU-CHI

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1 3.2-17

The pdf of chi-square distribution is:

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}$$

In this problem $r = 4$, so $f(x) = \frac{1}{4}xe^{-x/2}$. Let X be the r.v. of $\chi^2(4)$.

$$\begin{aligned} P(X > 7.779) &= 1 - P(X \leq 7.779) \\ &= 1 - \int_0^{7.779} \frac{1}{4}xe^{-x/2}dx \\ &\approx 1 - 0.9 = 0.1 \end{aligned}$$

Let Y be the number of observations (out of 15) that exceeds 7.779, since the observations are taken independently, it can be seen as a binomial distribution with $p = 0.1, n = 15$.

$$\begin{aligned} P(Y \leq 3) &= \sum_{k=0}^3 \binom{15}{k} (0.1)^k (0.9)^{15-k} \\ &= (0.9)^{15} + 15 \cdot (0.1)(0.9)^{14} + 105 \cdot (0.1)^2(0.9)^{13} + 455 \cdot (0.1)^3(0.9)^{12} \\ &= 0.9444 \end{aligned}$$

Hence, the probability that at most 3 of the 15 observations exceed 7.779 is 0.9444.

2 3.2-18

Let the r.v. $Y = X - 80$, so $Y > 0$ with pdf:

$$f(y) = \frac{y}{50^2} e^{-y/50}, \quad y > 0$$

This is Gamma distribution with shape $\alpha = 2$ and scale $\theta = 50$.

(a)

$$\begin{aligned}\mu_Y = E[Y] &= \alpha\theta = 2 \cdot 50 = 100 \\ \mu_X = E[X] &= E[Y + 80] = E[Y] + 80 = 180\end{aligned}$$

Shifting doesn't affect the variance:

$$\sigma_X^2 = \sigma_Y^2 = \alpha\theta^2 = 5000$$

Hence, the mean is 180 and the variance is 5000.

(b)

We can find the mode by setting $f'(y) = 0$:

$$\begin{aligned}f'(y) &= \frac{1}{2500}(e^{-y/50} - \frac{1}{50}ye^{-y/50}) \\ &= \frac{1}{2500}e^{-y/50}(1 - \frac{1}{50}y) = 0 \\ \Rightarrow y &= 50\end{aligned}$$

Hence, the mode of X is $50 + 80 = 130$.

(c)

This is equivalent to finding $P(Y < 120) = 1 - P(Y \geq 120)$:

$$\begin{aligned}P(Y < 120) &= 1 - \int_{120}^{\infty} \frac{y}{50^2} e^{-y/50} dy \\ &= 1 - \frac{1}{2500}(-50ye^{-y/50} \Big|_{120}^{\infty} + 50 \int_{120}^{\infty} e^{-y/50} dy) \\ &= 0.6915\end{aligned}$$

The percentage of having a serum cholesterol level less than 200 is 0.6915.

3 3.3-11

We are given that $\mu = 21.37, \sigma^2 = 0.16, \sigma = 0.4$.

(a)

Standardize using $Z = \frac{X-\mu}{\sigma}$.

$$P(X > 22.07) = P(Z > 1.75) = 1 - \Phi(1.75) = 1 - 0.9599 = 0.0401$$

where the value of $\Phi(1.75)$ is obtained by looking up the table.

We obtained that $P(X > 22.07) = 0.0401$

(b)

$$\begin{aligned} P(X < 20.857) &= P(Z < -1.28) \\ &= P(Z \leq -1.28) + P(Z = 1.28) \\ &= [1 - \Phi(1.28)] + \int_{1.28}^{1.28} f(x)dx \\ &= 1 - \Phi(1.28) = 1 - 0.8997 = 0.1003 \end{aligned}$$

(c)

Since each selection is independent, it can be seen as a binomial distribution with $p = 0.1$ and $n = 15$. (Y is the r.v.)

$$\begin{aligned} P(Y \leq 2) &= P(Y = 0) + P(Y = 1) + P(Y = 2) \\ &= \binom{15}{0}(0.9)^{15} + \binom{15}{1}(0.1)(0.9)^{14} + \binom{15}{2}(0.1)^2(0.9)^{13} \\ &= 0.8159 \end{aligned}$$

So we have that $P(Y \leq 2) = 0.8159$.

4 4.1-9

5 4.2-3